		Mistral-7B	PaLM-2 L	Compute
Methods without prompting	Greedy decoding	9.9%	34.8%	$O(1)$
	Self-consistency without CoT	12.9%	40.6%	$O(k)$
	CoT-decoding (max path)	25.1%	63.2%	$O(k)$
	CoT-decoding (agg path)	**25.3**%	**64.1**%	$O(k)$
Methods with prompting	Zero-shot CoT prompting	17.5%	75.1%	$O(1)$
	Self-consistency with zero-shot CoT-prompt	39.4%	85.3%	$O(k)$
	CoT-decoding (max path) + zero-shot CoT-prompt	40.2%	78.6%	$O(k)$
	CoT-decoding (agg path) + zero-shot CoT-prompt	**48.4%**	**87.0%**	$O(k)$

Table 7 | Adding CoT-decoding on top of zero-shot CoT-prompting can further boost the reasoning performance on both models. The accuracy number here is computed over the GSM8K test set.

4. Related Work

Chain-of-thought reasoning in large language models. Existing work enhancing the reasoning abilities in large language models predominantly involve proposing better prompting techniques to better elicit CoT reasoning paths from the model (Kojima et al., 2022; Nye et al., 2021; Wei et al., 2022; Yao et al., 2023; Yasunaga et al., 2023; Zhou et al., 2023a). Despite achieving high performance, few-shot prompting techniques are often *task-specific*, requiring prompt designs tailored to each task. This limits their generalizability across tasks. Advanced prompting techniques often require manually intensive prompt engineering, and their effectiveness varies depending on the choice of prompts, resulting in inconsistent performance outcomes (Wang et al., 2022; Ye and Durrett, 2022; Zhou et al., 2023b). Efforts to discover improved prompts (Yang et al., 2024; Zhou et al., 2023b) further entail model-specific and task-specific tuning.

In addition, these prompting techniques can subtly alter the vocabulary's posterior distribution in ways that remain largely elusive (Min et al., 2022; Webson and Pavlick, 2022). Specifically, prompts may assist in task decomposition, induce the model to generate additional tokens, or directly "teach" the model the exact underlying procedure to solve particular problems via manually crafted few-shot demonstrations. Dissecting the distinct influence of each aspect, however, presents a significant challenge. In contrast, our work explores a different perspective within the decoding stage, demonstrating that, even without explicit prompting, the model inherently holds the capability to generate chain-of-thought reasoning paths across a wide set of tasks.

Recent work proposes to improve the CoT generation process via better controlling and verifying the steps generated, e.g., step-by-step verification (Lightman et al., 2023), process-based feedback (Uesato et al., 2022), self-evaluation guided beam search (Xie et al., 2023), and PathFinder (Golovneva et al., 2023). Note all these works still require CoT prompting in order to generate the CoT reasoning paths, while our work completely removes CoT prompting. In addition, these works focus on searching and verifying the "steps" produced by the language model, while our work purely searches in the decoding space on the token-level and utilizes the confidence scores when decoding the answer.

Additionally, recent works (Feng et al., 2023; Li et al., 2023b; Prystawski et al., 2023). McCoy et al. (2023); Razeghi et al. (2022) demonstrate a similar phenomenon where the pretraining distribution heavily influences the model's performance in few-shot reasoning.

Instruction-tuning to elicit CoTs in language models. When supervision is allowed, techniques such as instruction-tuning or distillation offer another way to elicit reasoning paths from language models without explicit prompting (Chung et al., 2022; Huang et al., 2023; Magister et al., 2023). However, these approaches typically involve resource-intensive fine-tuning over large language models

and require a large set of examples annotated with CoTs, which may not be readily available.

Liu et al. (2024) show that a language model can be tuned by a proxy. Their method requires a few additional models, and implicitly assumes that the tuned model is well-optimized, e.g., on reasoning benchmarks the model needs to be tuned with CoT paths to enable contrasting logits with respect to the base untuned model. In contrast, our approach is entirely unsupervised and examines a model's intrinsic ability in generating CoT paths, without resorting to fine-tuning or any additional models.

Decoding algorithms for language models. The predominant focus in existing literature on decoding for language models revolves around aspects such as fluency, coherence, reduction of repetitiveness, and diversity in responses. Popular decoding algorithms used for language models include greedy decoding, temperature sampling (Ackley et al., 1985; Ficler and Goldberg, 2017), top-k sampling (Fan et al., 2018; Holtzman et al., 2018; Radford et al., 2019), and nucleus sampling (Holtzman et al., 2020). Additionally, there exist refined algorithms such as minimum Bayes risk decoding (Eikema and Aziz, 2020), and typical decoding (Meister et al., 2022). Diverse beam search (Vijayakumar et al., 2018) is another way to explore alternative paths in a model's generation. However, it emphasizes generation diversity rather than accuracy.

There is relatively little research dedicated to enhancing decoding algorithms specifically for reasoning tasks. Wang et al. (2023a) improves upon CoT prompting by sampling and aggregating over multiple generated responses to improve reasoning. Contrastive decoding (Li et al., 2023a) is another way to improve model's generation quality by penalizing the logits from smaller models, and recent work (O'Brien and Lewis, 2023) shows that contrastive decoding can contribute to enhancing reasoning performance. Shi et al. (2023) propose context-aware decoding to improves the faithfulness of language models. These approaches typically require additional information, such as employing additional models to generate contrasting logits or incorporating additional contexts. In contrast, our work relies solely on a single model without the need for supplementary knowledge.

Decoding algorithms for efficiency. In addition to decoding algorithms for improving quality, there is a substantial body of research dedicated to improving decoding efficiency, e.g., speculative decoding (Chen et al., 2023a; Leviathan et al., 2022; Zhou et al., 2024). This line of work is orthogonal to our work as their primary focus is not on improving a model's reasoning performance. However, these techniques could potentially be leveraged to improve the efficiency of CoT-decoding.

5. Conclusion and Discussion

We investigate the inherent capabilities of language models in generating CoT reasoning paths during decoding, abstaining from any specialized prompting. Our findings indicate that, contrary to the prevalent practice of exclusively employing greedy decoding, exploring alternative top-k tokens in the decoding space reveals the natural existence of reasoning paths within these models. Furthermore, our empirical observations highlight that the presence of a CoT reasoning path correlates with increased model confidence in decoding its final answer. Based on this observation, we introduce CoT-decoding to extract more reliable decoding paths from language models, thereby enhancing their overall reasoning performance.

Discussion and Limitations. The exploration of alternative decoding paths incurs additional computational costs. Future work could leverage the CoT-decoding paths to fine-tune the model to further enhance its reasoning capabilities. Additionally, in cases where the answers are more open-ended, utilizing the probability differences of the top two tokens as an indicator of how models prefer one answer over another could be less precise. While existing work (Burns et al., 2023)

leverages the model's activation space to uncover latent knowledge, its applicability is restricted to answering yes-no questions. We hope that future research can address this limitation by delving deeper into the model's internal representation across a broader, more open-ended answer space.

Furthermore, our current exploration focuses on branching at the first token, but for future work one can explore branching at any token and searching for the best possible paths during the decoding phase. The computational cost will be substantially higher though, and how to reliably identify the best token during the search will be an interesting direction to explore.

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